Total Number of Page-4 Roll No.----------

THIRD SEMESTER B. Tech.

Supplementary End-Semester Examination Feb-2019

HU- 201: Engineering Economics

Time: 3 Hour Max. Marks: 50

Note: Answer **Any Five** questions,
Assume suitable missing data, if any

1(a) Discuss in detail Price Elasticity and Cross Elasticity of Demand. Also discuss their relevance in business. 5

1(b) In the design of a jet engine part, the designer has a choice of specifying either an aluminum alloy casting or a steel casting. Either material will provide equal service, but the aluminum casting will weigh 1.2 Kg. as compared with 1.35 Kg. for steel casting. The aluminum can be cost Rs. 80.00 per Kg. and the steel one for Rs. 35.00 per Kg. The cost of machining per unit is Rs. 150.00 for aluminum and R. 170.00 for steel. Every Kilogram of excess weight is associated with a penalty of Rs. 1300 due to increased fuel consumption. Which material should be specified and what is the economic advantage of the selection per unit? 2

1(C) The chief engineer of refinery operation is not satisfied with the preliminary design for storage tanks to be used as part of a plant expansion programme. The engineer who submitted the design was called in and asked o reconsider the overall dimensions in the light of an article in the "Chemical Engineer", entitled "How to size future process vessels"? 3

The original design submitted called for 4 tanks 5.2 m is diameter and 7 m in height. From a graph of the article the engineer found that the present ratio of height to diameter of 1.35 is 111% of the minimum cost and that the minimum cost for a tank was when the ratio of height to

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diameter was 4:1. The cost for the tank design as originally submitted was estimated to be Rs. 9,00,000. What are the optimum tank dimensions if the volume remains the same as the original design? What total savings may be expected through the redesign?

2(a) If you are CEO of a company, discuss factors which you will consider while deciding price of your product in the market. 5

2(b) What amount of money saved today will yield Rs. 40,000/- in third year and Rs. 55,000/- after five year at the 12% rate of interest compounded annually. 2

2(c) A company has to replace a machine in the production line after 11 years at the cost of Rs. 60,00,000/-. It plans to deposit an equal amount at the end of every year for the next 11 years at an interest rate of 11 per cent which is compounded annually. Find the equivalent amount that must be deposited at the end of every year for next 11 years. 3

3(a) Discuss the nature of soft drink market in India. What incentive may be given by the Government to encourage Indian producer? 5

3(b) Below is given a demand equation; 2

$$Q = -6P + 4000$$

Calculate price elasticity of demand if price is (i) Rs. 4 (ii) Rs 10 and (iii) Rs. 15. Is the demand at these prices elastic or inelastic?

3(c) The market supply and demand functions for deluxe pizzas in a small town are given by 3

Demand $Q = 100 - 3.5\,P$

Supply $Q = 15 + 1.5\,P$

i. Determine the equilibrium price and quantity.

ii. If the city Government levies a tax of Rs. 3.00/pizza on the pizza parlor, determine the

new equilibrium price and quantity of pizza

4(a) What do you mean by Production Process? How it has changed in the era of Information and Communication Technology 5

4(b) A Company wants to deposit money to create an R&D reserve. The company will get 15,00,000 every year for next 15 years for R&D. The reserve will grow at the rate of 12 per cent annually. Find out the single payment which should be made now. 5

5(a) Environmental problems are macro level problems. Discuss the role of appropriate technology in this regard. Also suggest framework to impose taxes to control environmental hazards. 5

5(b) A firm, Prime Manufacturing is planning to expand its production operation. It has identified three machines which are technologically capable to serve the purpose. The initial outlay and annual revenues with each of the machines are given below: 5

	Initial Outlay (Rs.)	Annual revenue	Life (in years)
Machine I	Rs. 5,45,000	Rs. 2,50,000	15
Machine II	Rs. 6,14,000	Rs. 3,30,000	12
Machine III	Rs. 6,00,900	Rs. 3,50,000	10

If the rate of interest is 12%, which machine the company should opt for? Find out the result by Present Worth method.

the bids for the elevators are as follows:

Bids	Initial Cost(Rs.)	Service life(years)	Annual operations and maintenance cost(Rs.)
A elevator Inc.	4,500,000	15	27000
B elevator Inc.	5,400,000	15	28500

Determine which bid should be accepted based on the present worth method of comparison assuming 15% interest rate compounded annually. 7

[b] Explain the salient features of Indian economy. 7

5[a] You borrowed Rs. 21,061.82 to finance the educational expenses for your senior year college. The loan will be paid off over five years. The loan carries interest rate of 6% per year and is to repaid in equal installments over the next five years. Assume that the money was borrowed at the beginning of your senior year and that the first installment will be due a year later. Compute the amount of annual installments. 7

[b] What is the equal payment series for 10 years that is equivalent to a payment series starting with Rs. 20,000 at the end of the first year and decreasing by Rs. 2,000 each year over 10 years? Interest is 8% compounded annually. 7

6[a] You borrowed 10,000 from a local bank, with the agreement that you will payback the loan according to graduated payment plan. If your first payment is set at 1500, what would the remaining payment look like at a borrowing rate of 10% over five years.

[b] Consider the following two mutually exclusive alternatives.

	Alternative A	Alternative B
Cost(Rs.)	40,000	60,000
Uniform annual benefit(Rs.)	6400	9600
Useful life(years)	20	20

Using a 15% interest rate, determine which alternative should be selected based on the future Worth method of comparison.

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